Capstone: classification

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1 Data Preprocessing

I first handled missing values in the 'tempos' and 'duration_ms' by imputing them using the median of each column. And then I dropped several columns that are unrelated to the problem we are going to solve, like 'track_name', 'artist_name','instance_id','obtained_date'. I then transformed the string format variables 'key' into numerical values and 'music_genre' into numerical labels. To facilitate following analysis, I also dummy-coded the variable 'mode', scaled all the numerical values, and transformed the variable 'acousticness' using the boxcox method. The final dataset consists of $50,000$ samples and 13 features. By preprocessing the data in this way, I can ensure that the music data is of high quality and can be analyzed effectively to identify the factors that contribute to music genre.

After finishing all these, I did the train/test split as stated in the requirement. I tried LDA, PCA, t-SNE to perform dimensionality reduction on the data separately. For each method, I tried to keep different amounts of components to find the most appropriate number of components that explain at least 70-80% of the variance of the principal component in the data. The LDA turns out to be the one that can explain the most variance by the least number of components remained. Finally, I plot several solutions to visualize the separation between different genres.

Here is the 3D LDA result where these three components explain the about 92% variance.

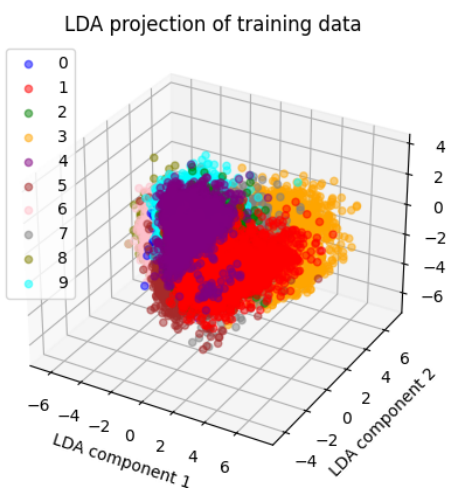


Figure 1: 3D LDA

From the figure, we can see that the clustering in the lower dimensional space is not well separated and that there is significant overlap between points in different clusters. This can make it more difficult for a classification model to accurately predict the music genre based on its features.

2 Classificaiton Model

I chose three kinds of classification models: Random Forest, SVM, and Neural Networks. To evaluate the performance of the models, I plotted an ROC curve and calculated the AUC score for each genre. This helped me understand the ability of the models to distinguish between different genres.

I chose these three models because they have been shown to perform well in classification tasks and are commonly used in machine learning applications. Additionally, by choosing a mix of linear and non-linear models, I hoped to cover a range of possible relationships between the input features and the output labels. The reason for calculating the ROC curves and AUC for each class separately to evaluate the performance of the classification models predicting multiple classes is that each class has its own distribution of true and false positives and negatives, and evaluating them separately provides a more detailed understanding of the model's performance.

3 Results

3.1 Random Forest

Random Forest is an ensemble learning method that builds multiple decision trees and aggregates their predictions to achieve better performance. The result of the ROC and AUC of each genre class by using the LDA data is shown in the figure (a). By calculation, the weighted AUC in this situation is about 87.3%.

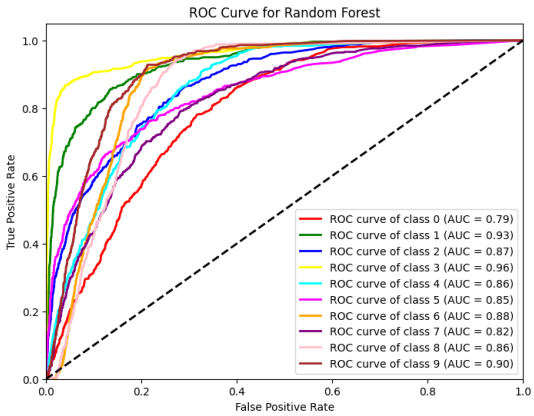
3.2 SVM

SVM, on the other hand, constructs a hyperplane that separates the classes in a high-dimensional space, with the aim of maximizing the margin between the classes. In the case of using LDA data, the weighted AUC is about 83.7%.

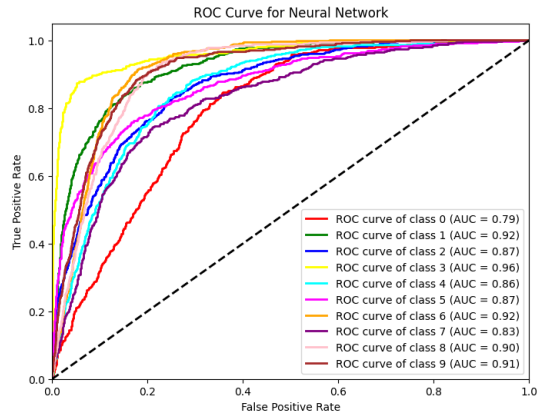
3.3 Feedforward Neural Network

Neural Networks are a class of models that are particularly well-suited for handling complex, non-linear relationships between the input features and the output labels.

I tried two feedforward neural networks with two hidden layers, one uses the activation function reLU and another uses sigmoid function. The result of the ROC and AUC of each genre class for the reLU neural network model when using the LDA data is shown in figure (b). The weighted AUC is 88.3%.



(a) ROC curve for random forest



(b) ROC curve for neural network

4 Conclusion

The highest AUC I got is 88.3%, which indicates good classification performance. The selection of the dimensionality reduction strategy and hyperparameter is what contributes most to the success of my classification. The LDA, PCA, and t-SNE all yield significantly different AUC scores (especially t-SNE), demonstrating the importance of selecting a dimensionality reduction that is appropriate for the dataset. Meanwhile, different hyperparameters will cause very different output. For example, the epoch value in the neural network model would greatly impact the final AUC score.

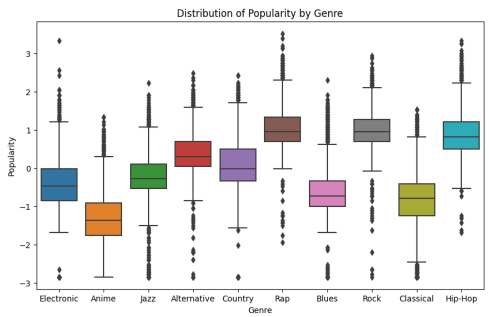
5 Interesting Observations

5.1 a

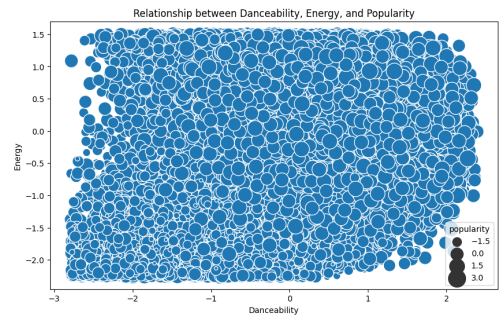
In the box plot of popularity by music genre below, the boxes show the distribution of the popularity scores for each music genre. From the figure, we can see that the music genres like pop, rock, and hip-hop tend to have more popular songs than less popular songs. And music genres like anime, blues, and classical tend to have less popular songs than more popular songs. Besides, the Jazz music genre has many outliers above and below the box, which means that the popularity values for Jazz songs vary widely, with some songs being very popular and others not so much.

5.2 b

I created a scatter plot that shows the relationship between a song's danceability and energy, with the size of each point on the plot representing the song's popularity. As can be seen from the figure, when the danceability and energy are low, the popularity tends to be low and when both dance and energy increase, popularity increases. But when both dancing ability and energy reach a certain value, popularity does not change much.



(c) relationship between popularity and music genres



(d) relationship between danceability, energy, and popularity